

Planning Wizard - Calculation Formulas

This document contains all the formulas used in the Planning Wizard for technical sizing, economic analysis, and emissions calculations.

Technical Recommended Rating FORMULAS

1.1 Basic Power Calculations

Average Power (kW)

$$\text{Average Power} = \text{Daily Consumption (kWh)} / 24 \text{ hours}$$

Peak Power (kW)

$$\text{Peak Power} = \text{Average Power} \times \text{Peak-to-Average Ratio}$$

Where Peak-to-Average Ratio:

- Residential: 2.5
- Commercial: 1.8
- Industrial: 1.3

Annual Consumption (kWh)

$$\text{Annual Consumption} = \text{Daily Consumption} \times 365 \text{ days}$$

1.2 Solar Capacity Sizing

Base Solar Capacity (kW)

$$\text{Base Solar Capacity} = \text{Daily Consumption} / (\text{Peak Sun Hours} \times \text{System Efficiency})$$

Where:

- Peak Sun Hours = 5 hours/day
- System Efficiency = 0.773 (77.3%)

Primary Goal Multipliers (Applied to Base Solar Capacity):

$$\text{Solar Capacity} = \text{Base Solar Capacity} \times \text{Solar Multiplier}$$

Where Solar Multiplier:

- Cost Savings: 0.9 (10% reduction to minimize initial investment)
- Self-Sustainability: 1.3 (30% increase to maximize renewable energy)
- Reliability: 1.1 (10% increase for consistent power generation)
- Carbon Reduction: 1.2 (20% increase to minimize grid emissions)
- Default (if no goal specified): 1.0

Daily Solar Generation (kWh)

$$\text{Daily Solar Generation} = \text{Solar Capacity} \times \text{Peak Sun Hours} \times \text{System Efficiency}$$

1.3 Battery Capacity Sizing

Base Battery Capacity (kWh)

$$\text{Base Battery Capacity} = \text{Daily Consumption} \times \text{Base Battery-to-Daily Ratio}$$

Where Base Battery-to-Daily Ratio:

- Residential: 0.83 (83%)
- Commercial: 0.65 (65%)
- Industrial: 0.55 (55%)

Primary Goal Multipliers (Applied to Base Battery Capacity):

$$\text{Battery Capacity} = \text{Base Battery Capacity} \times \text{Battery Multiplier}$$

Where Battery Multiplier:

- Cost Savings: 0.8 (20% reduction to minimize initial investment)
- Self-Sustainability: 1.3 (30% increase to maximize renewable energy storage)
- Reliability: 1.5 (50% increase for extended backup duration)
- Carbon Reduction: 1.2 (20% increase to reduce grid dependence)
- Default (if no goal specified): 1.0

Battery Capacity (Ah)

$$\text{Battery Capacity (Ah)} = (\text{Battery Capacity (kWh)} \times 1000) / \text{Battery Voltage (V)}$$

Where Battery Voltage:

- Residential: 12V
- Commercial: 48V
- Industrial: 480V

Backup Hours

$$\text{Backup Hours} = \text{Battery Capacity (kWh)} / \text{Average Power (kW)}$$

1.4 Inverter & DC-DC Converter Sizing

Inverter Capacity (kW/kVA)

$$\text{Inverter Capacity} = \text{Solar Capacity} \times 1.25$$

DC-DC Converter Rating (kW)

$$\text{DC-DC Converter Rating} = \text{Solar Capacity} \times 1.25$$

1.5 Grid & Diesel Sizing

Grid Connection (kW)

$$\text{Grid Connection} = \text{Peak Power (kW)}$$

Diesel Capacity (kW)

$$\text{Diesel Capacity} = \text{Peak Power} \times 1.1$$

1.6 Hydrogen System (Optional)

Electrolyzer Capacity (kW)

Electrolyzer Capacity = Solar Capacity × 0.2 (20% of solar capacity)

Purpose: Converts excess solar energy to hydrogen for long-duration storage

Fuel Cell Capacity (kW)

Fuel Cell Capacity = Peak Power × 0.5 (50% of peak power)

Purpose: Provides backup power generation from stored hydrogen

Note: This rating is displayed in Technical Analysis when hydrogen is enabled

H2 Tank Capacity (kg)

H2 Tank Capacity = Battery Capacity (kWh) / 33.3

Where: 1 kg H2 ≈ 33.3 kWh (energy equivalent)

Purpose: Stores hydrogen for fuel cell operation during extended outages

Hydrogen System Cost Calculation:

Electrolyzer Cost = Electrolyzer Capacity × ₹50,000

Fuel Cell Cost = Fuel Cell Capacity × ₹60,000

H2 Tank Cost = H2 Tank Capacity × ₹5,000

Total Hydrogen CAPEX = Electrolyzer Cost + Fuel Cell Cost + H2 Tank Cost

Display in Technical Analysis: When hydrogen system is enabled, the following fields are shown:

- Fuel Cell Rating (kW): Fuel Cell Capacity
- Electrolyzer Rating (kW): Electrolyzer Capacity
- H2 Tank Capacity (kg): H2 Tank Capacity

2. ECONOMIC ANALYSIS FORMULAS

2.1 Cost Assumptions

Unit Costs:

- Solar Cost: ₹40,000 per kW
- Battery Cost: ₹8,000 per kWh
- Inverter Cost: ₹8,000 per kW
- DC-DC Converter Cost: ₹1,500 per kW
- Installation: 10% of equipment cost
- O&M: 3% of (Solar + Battery) cost per year

Hydrogen System Costs (if applicable):

- Electrolyzer: ₹50,000 per kW
- Fuel Cell: ₹60,000 per kW
- H2 Tank: ₹5,000 per kg

2.2 Dual Mode System Costs

Equipment Costs:

Solar Cost = Solar Capacity × ₹40,000

Battery Cost = Battery Capacity × ₹8,000

Inverter Cost = Inverter Capacity × ₹8,000
DC-DC Converter Cost = DC-DC Converter Rating × ₹1,500

Installation Cost:

Installation Cost = (Solar + Battery + Inverter + DC-DC Converter) × 0.10

Total CAPEX:

Total CAPEX = Equipment Cost + Installation Cost

Annual O&M Cost:

Annual O&M = (Solar Cost + Battery Cost) × 0.03
Monthly O&M = Annual O&M / 12

2.3 On-Grid System Costs

Equipment Costs:

Solar Cost = Solar Capacity × ₹40,000
Inverter Cost = Inverter Capacity × ₹8,000
Battery Cost = 0 (No battery)
DC-DC Converter Cost = 0 (No DC-DC converter)

Installation Cost:

Installation Cost = (Solar + Inverter) × 0.10

Total CAPEX:

Total CAPEX = Equipment Cost + Installation Cost

Annual O&M Cost:

Annual O&M = Solar Cost × 0.03
Monthly O&M = Annual O&M / 12

2.4 Annual Generation & Revenue

System Efficiencies:

- Dual Mode System Efficiency: 0.80 (80%) - lower due to battery round-trip losses
- On-Grid System Efficiency: 0.85 (85%) - higher, direct grid connection

Annual Solar Generation:

Dual Mode: Annual Generation = Solar Capacity × 5 × 365 × 0.80
On-Grid: Annual Generation = Solar Capacity × 5 × 365 × 0.85

Grid Tariffs:

- Grid Export Tariff: ₹8.0 per kWh (selling excess solar)
- Grid Purchase Tariff: ₹8.5 per kWh (buying from grid)

Annual Revenue:

Dual Mode Revenue = Annual Generation (Dual Mode) × ₹8.0
On-Grid Revenue = Annual Generation (On-Grid) × ₹8.0

Grid Energy Needed:

Dual Mode Grid Energy = max(0, Annual Consumption – Annual Generation (Dual Mode))
On-Grid Grid Energy = max(0, Annual Consumption – Annual Generation (On-Grid))

Annual Grid Cost:

Dual Mode Grid Cost = Dual Mode Grid Energy × ₹8.5
On-Grid Grid Cost = On-Grid Grid Energy × ₹8.5

2.5 Annual Savings & Payback Period

Annual Savings:

Dual Mode Savings = Annual Revenue – Annual Grid Cost – Annual O&M
On-Grid Savings = Annual Revenue – Annual Grid Cost – Annual O&M

Monthly Savings:

Monthly Savings = Annual Savings / 12

Payback Period (years):

Payback Period = Total CAPEX / Annual Savings

(If Annual Savings ≤ 0, Payback Period = 0)

3. CARBON EMISSIONS CALCULATION FORMULAS

3.1 Emission Factors

Emission Factors:

- Grid Emission Factor: 0.82 kg CO₂ per kWh (Indian grid average)
- Solar Operational Factor: 0.05 kg CO₂ per kWh (operational emissions)
- Solar Manufacturing Factor: 0.04 kg CO₂ per kWh capacity (manufacturing)
- Battery Manufacturing Factor: 150 kg CO₂ per kWh capacity (manufacturing)
- System Lifetime: 25 years

3.2 Lifetime Emissions Calculation

Lifetime Consumption:

Lifetime Consumption = Annual Consumption × 25 years

Dual Mode System - Lifetime Emissions (25 years):

1. Solar Manufacturing Emissions (one-time):

Solar Manufacturing = Solar Capacity (kW) × 0.04 × 1000

2. Solar Operational Emissions:

$\text{Solar Operational} = \text{Annual Solar Generation (Dual Mode)} \times 0.05 \times 25 \text{ years}$

3. Grid Emissions:

$\text{Grid Emissions} = \text{Dual Mode Grid Energy Needed} \times 0.82 \times 25 \text{ years}$

4. Battery Manufacturing Emissions (one-time):

$\text{Battery Manufacturing} = \text{Battery Capacity (kWh)} \times 150$

5. Total Dual Mode Lifetime Emissions:

$\text{Total Emissions (kg)} = \text{Solar Manufacturing} + \text{Solar Operational} + \text{Grid Emissions} + \text{Battery Manufacturing}$

$\text{Total Emissions (tonnes)} = \text{Total Emissions (kg)} / 1000$

On-Grid System - Lifetime Emissions (25 years):

1. Solar Manufacturing Emissions (one-time):

$\text{Solar Manufacturing} = \text{Solar Capacity (kW)} \times 0.04 \times 1000$

2. Solar Operational Emissions:

$\text{Solar Operational} = \text{Annual Solar Generation (On-Grid)} \times 0.05 \times 25 \text{ years}$

3. Grid Emissions:

$\text{Grid Emissions} = \text{On-Grid Grid Energy Needed} \times 0.82 \times 25 \text{ years}$

4. Total On-Grid Lifetime Emissions:

$\text{Total Emissions (kg)} = \text{Solar Manufacturing} + \text{Solar Operational} + \text{Grid Emissions}$

$\text{Total Emissions (tonnes)} = \text{Total Emissions (kg)} / 1000$

3.3 Annual Emissions (for display)

Annual Emissions:

$\text{Dual Mode Annual Emissions} = \text{Dual Mode Lifetime Emissions} / 25$

$\text{On-Grid Annual Emissions} = \text{On-Grid Lifetime Emissions} / 25$

3.4 Carbon Offset Calculation

Pure Grid Baseline:

$\text{Pure Grid Annual Emissions} = \text{Annual Consumption} \times 0.82$

$\text{Pure Grid Lifetime Emissions} = \text{Lifetime Consumption} \times 0.82$

Dual Mode Operational Emissions (Annual):

$\text{Dual Mode Operational Annual} = (\text{Solar Operational} + \text{Grid Emissions}) / 25 \text{ years}$

Note: Excludes manufacturing emissions for carbon offset calculation

Annual CO₂ Reduction:

Annual CO₂ Reduction = Pure Grid Annual Emissions – Dual Mode Operational Annual Emissions

Carbon Offset Percentage:

Carbon Offset % = (Annual CO₂ Reduction / Pure Grid Annual Emissions) × 100

Lifetime CO₂ Reduction:

Lifetime CO₂ Reduction (tonnes) = (Pure Grid Lifetime Emissions – Dual Mode Lifetime Emissions) / 1000

4. PRIMARY GOAL OPTIMIZATION STRATEGY

The system adjusts solar and battery sizing based on the selected primary goal to optimize for different objectives:

4.1 Cost Savings Optimization

Strategy: Minimize initial investment while maintaining basic functionality

- Solar: 10% reduction (multiplier: 0.9)
- Battery: 20% reduction (multiplier: 0.8)
- **Result:** Lower CAPEX, faster payback, but reduced self-sufficiency

4.2 Self-Sustainability Optimization

Strategy: Maximize renewable energy usage and independence from grid

- Solar: 30% increase (multiplier: 1.3)
- Battery: 30% increase (multiplier: 1.3)
- **Result:** Higher renewable energy generation, better energy independence, longer backup duration

4.3 Reliability Optimization

Strategy: Ensure continuous power supply with extended backup duration

- Solar: 10% increase (multiplier: 1.1)
- Battery: 50% increase (multiplier: 1.5)
- **Result:** Extended backup hours, better resilience during outages, higher reliability

4.4 Carbon Reduction Optimization

Strategy: Minimize grid emissions through increased renewable energy

- Solar: 20% increase (multiplier: 1.2)
- Battery: 20% increase (multiplier: 1.2)
- **Result:** Lower carbon footprint, reduced grid dependence, balanced cost-performance

5. EXAMPLE CALCULATIONS

Example 1: Base Calculation (No Primary Goal)

Input:

- Daily Consumption: 9.90 kWh/day
- Use Case: Residential
- Primary Goal: None (default)

Step 1: Technical Sizing

Average Power = $9.90 / 24 = 0.4125$ kW
Peak Power = $0.4125 \times 2.5 = 1.03125$ kW
Base Solar Capacity = $9.90 / (5 \times 0.773) = 2.56$ kW
Solar Capacity = $2.56 \times 1.0 = 2.56$ kW (no multiplier)
Base Battery Capacity = $9.90 \times 0.83 = 8.217$ kWh
Battery Capacity = $8.217 \times 1.0 = 8.21$ kWh (no multiplier)
Inverter Capacity = $2.56 \times 1.25 = 3.2$ kW ≈ 3.19 kW
DC-DC Converter = $2.56 \times 1.25 = 3.2$ kW ≈ 3.19 kW

Example 2: Cost Savings Goal

Input:

- Daily Consumption: 9.90 kWh/day
- Use Case: Residential
- Primary Goal: Cost Savings

Step 1: Technical Sizing

Average Power = $9.90 / 24 = 0.4125$ kW
Peak Power = $0.4125 \times 2.5 = 1.03125$ kW
Base Solar Capacity = $9.90 / (5 \times 0.773) = 2.56$ kW
Solar Capacity = $2.56 \times 0.9 = 2.304$ kW (10% reduction)
Base Battery Capacity = $9.90 \times 0.83 = 8.217$ kWh
Battery Capacity = $8.217 \times 0.8 = 6.574$ kWh ≈ 6.57 kWh (20% reduction)
Inverter Capacity = $2.304 \times 1.25 = 2.88$ kW
DC-DC Converter = $2.304 \times 1.25 = 2.88$ kW

Example 3: Reliability Goal

Input:

- Daily Consumption: 9.90 kWh/day
- Use Case: Residential
- Primary Goal: Reliability

Step 1: Technical Sizing

Average Power = $9.90 / 24 = 0.4125$ kW
Peak Power = $0.4125 \times 2.5 = 1.03125$ kW
Base Solar Capacity = $9.90 / (5 \times 0.773) = 2.56$ kW
Solar Capacity = $2.56 \times 1.1 = 2.816$ kW (10% increase)
Base Battery Capacity = $9.90 \times 0.83 = 8.217$ kWh
Battery Capacity = $8.217 \times 1.5 = 12.326$ kWh ≈ 12.33 kWh (50% increase)
Backup Hours = $12.33 / 0.4125 = 29.9$ hours (extended backup)
Inverter Capacity = $2.816 \times 1.25 = 3.52$ kW
DC-DC Converter = $2.816 \times 1.25 = 3.52$ kW

Example 4: With Hydrogen System

Input:

- Daily Consumption: 9.90 kWh/day
- Use Case: Residential
- Primary Goal: Self-Sustainability
- Include Hydrogen: Yes

Step 1: Technical Sizing

Base Solar Capacity = $9.90 / (5 \times 0.773) = 2.56 \text{ kW}$
Solar Capacity = $2.56 \times 1.3 = 3.328 \text{ kW}$ (30% increase for self-sustainability)
Base Battery Capacity = $9.90 \times 0.83 = 8.217 \text{ kWh}$
Battery Capacity = $8.217 \times 1.3 = 10.682 \text{ kWh}$ (30% increase)
Peak Power = $0.4125 \times 2.5 = 1.03125 \text{ kW}$

Step 2: Hydrogen System Sizing

Electrolyzer Capacity = $3.328 \times 0.2 = 0.666 \text{ kW}$
Fuel Cell Capacity = $1.03125 \times 0.5 = 0.516 \text{ kW}$
H2 Tank Capacity = $10.682 / 33.3 = 0.321 \text{ kg}$

Step 3: Hydrogen System Costs

Electrolyzer Cost = $0.666 \times ₹50,000 = ₹33,300$
Fuel Cell Cost = $0.516 \times ₹60,000 = ₹30,960$
H2 Tank Cost = $0.321 \times ₹5,000 = ₹1,605$
Total Hydrogen CAPEX = $₹33,300 + ₹30,960 + ₹1,605 = ₹65,865$

Example 5: Economic Analysis (Cost Savings Goal)

Input:

- Daily Consumption: 9.90 kWh/day
- Use Case: Residential
- Primary Goal: Cost Savings
- Solar Capacity: 2.304 kW (from Example 2)
- Battery Capacity: 6.574 kWh (from Example 2)

Step 1: Equipment Costs

Solar Cost = $2.304 \times ₹40,000 = ₹92,160$
Battery Cost = $6.574 \times ₹8,000 = ₹52,592$
Inverter Cost = $2.88 \times ₹8,000 = ₹23,040$
DC-DC Converter Cost = $2.88 \times ₹1,500 = ₹4,320$
Equipment Total = ₹172,112

Step 2: Installation & CAPEX

Installation = $₹172,112 \times 0.10 = ₹17,211.20$
Total CAPEX = $₹172,112 + ₹17,211.20 = ₹189,323.20$

Comparison with Reliability Goal:

- Cost Savings CAPEX: ₹189,323.20
- Reliability CAPEX (from Example 3): Higher due to larger battery
- **Savings:** Cost Savings goal reduces CAPEX by optimizing for lower initial investment

Example 6: Annual Generation Comparison

Input: Daily Consumption: 9.90 kWh/day

Cost Savings Goal:

Solar Capacity = 2.304 kW
Annual Generation = $2.304 \times 5 \times 365 \times 0.80 = 3,363.84 \text{ kWh}$

Self-Sustainability Goal:

Solar Capacity = 3.328 kW (from Example 4)

Annual Generation = $3.328 \times 5 \times 365 \times 0.80 = 4,854.88$ kWh

Result: Self-Sustainability generates 44% more energy, demonstrating the optimization strategy

Last Updated: Based on `backend/routes/planning.js` implementation (Updated with Primary Goal Optimization - 2025)